Project Proposal

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1 Introduction

In this project we will be exploring Syntax Aware classification to determine the impact of syntax information through embeddings, parsing and specialized datasets, like the Stanford Sentiment Treebank, on sentiment analysis.

2 Summary of relevant research papers

For this project we have looked at different research papers. Now we will summarize the ones we have found most relevant.

The main one is the Stanford's "Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank" [5]. This paper explains the creation of the "Stanford Sentiment Treebank", a dataset aimed to improve the accuracy of sentiment analysis's classification. During the research they've created multiple models, from recursive models to their own Recursive Neural Tensor Network (RNTN), to evaluate the impact of the dataset. They found an increase in the accuracy, especially with the use of the RNTN. From this paper, we will use the dataset, as we want to evaluate the impact such datasets have on classification, as well some of the models displayed and their working methodology.

Another paper we found interesting is "Opinion Tree Parsing for Aspect-based Sentiment Analysis" [2]. Their objective is to reduce the time and resources needed for generative models in aspect base sentiment analysis as well as use the phrase structure to help the model determine the underlying sentiment. To achieve this, they've created an opinion tree parsing model that has had some promising results with a high accuracy and lower computational needs. The key idea we hope to use from this paper is the use of parsing trees in a model, in other words, determining how does a model for sentiment analysis work if the training data is a parsing tree.

Related to this one, there is "Combination of Recursive and Recurrent Neural Networks for Aspect-Based Sentiment Analysis Using Inter-Aspect Relations" [1]. This paper aims to create an ensemble model using recurrent and recursive models to improve state-of-the-art accuracy in sentiment analysis. Their model first uses recursive trees to extract syntactic structures from the phrase and then feeds those trees to a recurrent model. This ensemble was found to be a comprehensive and robust model. With this paper, we hope to deeply comprehend recursive models to implement one.

Another paper is "Dependency Based Embeddings for Sentence Classification Tasks" [3]. They compare the accuracy of a classification task using three types of embeddings, combining traditional skipgram with dependency parsing. In the three cases the best model was the one using dependency parsing in their embeddings. The key idea we have extracted from this paper is how to combine embeddings with a dependency graph.

We have also found relevant the paper "Dependency-Based Word Embeddings" [4]. In this paper, the authors describe how embeddings change when created by two variations of skipgram (traditional SGNS and a combination with dependency parsing). The results showed key differences in the final representations, mostly related to semantic versus syntactically similar words. From this one we can extract the variations to the embeddings for better representation.

3 Milestones and project plan

The plan for the project is the following: first we will work with a recursive model with the SST dataset. Once this basic model works properly, we will compare it with RNN, Naive Bayes and VecAvg, trained with the same dataset. At this point we will try to replicate the RNTN and compare it to the previously mentioned models. Finally, if we have time, we will train another model without the parsing to know the real difference parsing makes. We will evaluate the results mainly measuring the accuracy, although we may also consider the training cost in time and computational resources.

We have explored the SST dataset and are currently investigating how to utilize it with a recursive model. We have also thoroughly analysed the "Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank" paper to get a better understanding of both the dataset and the models used. In terms of coding, we have made a basic parsing implementation with the Penn dataset to familiarize ourselves with parsing trees. Furthermore, we are studying the recursive models papers to understand how to implement the first basic model and we have obtained word embeddings from BERT.

For April 12th, we hope to have a basic implementation of a recursive model where it is easy to see the steps the model does with the tree and also the beginning of Naive Bayes and VecAvg. For April 19th, we hope to have completed the implementations of all the models, including the implementation of a recurrent model (maybe bidirectional) and RNTN, to be able to adjust the hyperparameters and have the models trained by April 22nd.

References

[1] Cem Rifki Aydin and Tunga Güngör. "Combination of Recursive and Recurrent Neural Networks for Aspect-Based Sentiment Analysis Using Inter-Aspect Relations". In: *IEEE Access* 8 (2020), pp. 77820–77832. DOI: `10.1109/ACCESS.2020.2990306`.

[2] Xiaoyi Bao. *Opinion Tree Parsing for Aspect-based Sentiment Analysis*. 2023. URL: `https://aclanthology.org/2023.findings-acl.505.pdf`.

[3] Alexandros Komninos. *Dependency Based Embeddings for Sentence Classification Tasks*. 2016. URL: `https://aclanthology.org/N16-1175.pdf`.

[4] Omer Levy. *Dependency-Based Word Embeddings*. 2014. URL: `https://aclanthology.org/P14-2050/`.

[5] Richard Socher. *Recursive Deep Models for Semantic Compositionality Over a Sentiment Treebank*. 2013. URL: `https://nlp.stanford.edu/~socherr/EMNLP2013_RNTN.pdf`.